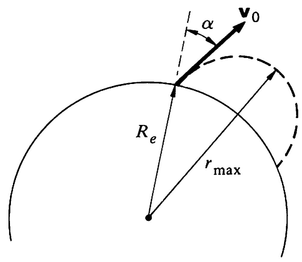
University Physics Quiz 3

Name: ______________________ SID: ________________________ School/Department: __________________

1. A projectile of mass m is launched from the surface of the earth at an angle α to the vertical. The initial speed is $v_0 = \sqrt{GM_e/R_e}$ where M_e and R_e are the mass and radius of the earth respectively. How high does the projectile rise? (The resistance of the air and the rotation of the earth can be neglected.)



Solution: The conservation of the angular momentum reads

$$mR_e v_0 \sin\alpha = mvr_{\max} . \qquad (1)$$

The conservation of mechanical energy reads

$$\frac{1}{2}mv_0^2 - \frac{GM_e m}{R_e} = \frac{1}{2}mv^2 - \frac{GM_e m}{r_{\max}} . \qquad (2)$$

From Eq.(1), we have $v = \dfrac{v_0 R_e \sin\alpha}{r_{\max}}$. Substituting $v = \dfrac{v_0 R_e \sin\alpha}{r_{\max}}$ into Eq.(2) we obtain

$$\frac{1}{2}v_0^2 - \frac{GM_e}{R_e} = \frac{1}{2}v_0^2\frac{R_e^2}{r_{\max}^2}\sin^2\alpha - \frac{GM_e}{r_{\max}} .$$

Noting that $v_0^2 = \dfrac{GM_e}{R_e}$, we obtain

$$-\frac{1}{2}\frac{GM_e}{R_e} = \frac{1}{2}\frac{GM_e}{R_e}\frac{R_e^2}{r_{\max}^2}\sin^2\alpha - \frac{GM_e}{r_{\max}} \quad \text{or} \quad \sin^2\alpha\left(\frac{R_e}{r_{\max}}\right)^2 - 2\frac{R_e}{r_{\max}} + 1 = 0 .$$

Then,

$$\frac{R_e}{r_{\max}} = \frac{2 \pm \sqrt{4 - 4\sin^2\alpha}}{2\sin^2\alpha} = \frac{1 \pm \cos\alpha}{\sin^2\alpha}$$

Therefore, (noting that $r_{\max} > R_e$)

$$\frac{r_{\max}}{R_e} = \frac{\sin^2\alpha}{1-\cos\alpha} \quad \text{or} \quad r_{\max} = \frac{\sin^2\alpha}{1-\cos\alpha}R_e$$

2. A satellite is moving on an elliptic orbit, and its perigee values of speed and distance from the center of the earth are v_p and r_p respectively. M_E is the mass of the earth.

(a) Show that the eccentricity of the orbit is given by $e = r_p v_p^2 / GM_E - 1$.

(b) Suppose that at the perigee of the elliptic orbit, the satellite experiences a drag due to the earth's upper atmosphere which has the effect of reducing its speed from v_p to $v_p(1-\varepsilon)$, where $0 < \varepsilon << 1$. Show that the eccentricity of its new orbit is approximately given by $e_1 = e - 2\varepsilon(1+e)$, where e is the original eccentricity of the orbit.

Solution: (a) According to the Eq.(4.4.19) of the textbook, the equation of orbit can be written as

$$r = \frac{p}{1 + e\cos\theta}$$

where e is the eccentricity. For an elliptic orbit, $0 < e < 1$ and perigee and apogee distances for the center of the earth are

$$r_p = \frac{p}{1+e}, \quad r_a = \frac{p}{1-e}.$$

Thus, the major axis is

$$2a = r_p + r_a = \frac{2p}{1-e^2}. \tag{1}$$

We also have

$$r_p = (1-e)a.$$

According to the Eq. (4.4.18) of the textbook, the energy of the satellite is

$$.E = -(1-e^2)\frac{GM_E m}{2p}.$$

Using (1), we have

$$E = -\frac{GM_E m}{2a}.$$

The conservation of energy gives

$$E = \frac{1}{2}mv_p^2 - \frac{GM_E m}{r_p} = -\frac{GM_E m}{2a} = -(1-e)\frac{GM_E m}{2r_p}.$$

Thus

$$v_p^2 = GM_E\left(\frac{2}{r_p} - (1-e)\frac{1}{r_p}\right) = \frac{GM_E}{r_p}(1+e).$$

Finally

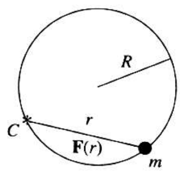
$$e = \frac{r_p v_p^2}{GM_E} - 1.$$

(b)

$$e_1 = \frac{r_p v_p^2 (1-\varepsilon)^2}{GM_E} - 1 = \frac{r_p v_p^2 (1 - 2\varepsilon + \varepsilon^2)}{GM_E} - 1 \approx \frac{r_p v_p^2 (1-2\varepsilon)}{GM_E} - 1$$

$$= \frac{r_p v_p^2}{GM_E} - 1 - 2\varepsilon\frac{r_p v_p^2}{GM_E} = e - 2\varepsilon(1+e).$$

3. A particle of mass m moves in a circular orbit of radius R under the influence of a central force $F(r)$. The center of the central force lies at a point on the circle, as shown in the accompanying figure. What is the force law of the central force? (Hint: the equation of the orbit can be written as $r = 2R\cos\theta$.)



Solution1: The equation of motion of the particle is

$$m\ddot{\boldsymbol{r}} = F(r)\boldsymbol{e}_r.$$

Noting that in planar polar coordinates $\ddot{\boldsymbol{r}} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\boldsymbol{e}_\theta$, the equation of motion can be written in the component form as

$$m(\ddot{r} - r\dot{\theta}^2) = F(r), \tag{1}$$

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0. \tag{2}$$

The integrating of the second equation will lead to

$$mr^2\dot{\theta} = L = \text{Const.} \tag{3}$$

Using the trajectory equation $r = r(\theta) = 2R\cos\theta$, and considering $r = r(\theta(t))$, we have

$$\dot{r} = \frac{dr}{d\theta}\frac{d\theta}{dt} = \frac{dr}{d\theta}\dot{\theta},$$

$$\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d}{dt}\left(\frac{dr}{d\theta}\dot{\theta}\right) = \frac{d}{dt}\left(\frac{dr}{d\theta}\right)\dot{\theta} + \frac{dr}{d\theta}\frac{d}{dt}(\dot{\theta}) = \frac{d}{d\theta}\left(\frac{dr}{d\theta}\right)\frac{d\theta}{dt}\dot{\theta} + \frac{dr}{d\theta}\ddot{\theta} = \frac{d^2r}{d\theta^2}\dot{\theta}^2 + \frac{dr}{d\theta}\ddot{\theta}.$$

From (2), we have $\ddot{\theta} = -\dfrac{2\dot{r}\dot{\theta}}{r} = -2\dfrac{dr}{d\theta}\dot{\theta}^2/r$. From (3), we have $\dot{\theta} = \dfrac{L}{mr^2}$. Therefore

$$F(r) = m(\ddot{r} - r\dot{\theta}^2) = -m\left(\frac{d^2r}{d\theta^2}\dot{\theta}^2 + \frac{dr}{d\theta}\ddot{\theta} - r\dot{\theta}^2\right) =$$

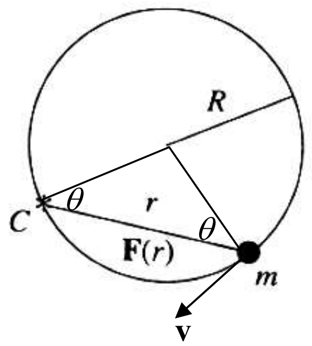
$$= m\left[\frac{d^2r}{d\theta^2}\dot{\theta}^2 - 2\left(\frac{dr}{d\theta}\right)^2\frac{\dot{\theta}^2}{r} - r\dot{\theta}^2\right] = -m\dot{\theta}^2\left[\frac{d^2r}{d\theta^2} - 2\left(\frac{dr}{d\theta}\right)^2\frac{1}{r} - r\right]$$

$$= m\left(\frac{L}{mr^2}\right)^2\left[-2R\cos\theta - 2\frac{(-2R\sin\theta)^2}{2R\cos\theta} - 2R\cos\theta\right]$$

$$= \frac{L^2}{mr^4}\left[-4R\cos\theta - 4\frac{R(1-\cos^2\theta)}{\cos\theta}\right] = -\frac{L^2}{mr^4}\frac{4R}{\cos\theta} = -\frac{8R^2L^2}{mr^4}\frac{1}{2R\cos\theta} = -\frac{8R^2L^2}{mr^5}$$

Solution2: The equation of the circular orbit can be written as

$$r = 2R\cos\theta, \quad \cos\theta = \frac{r}{2R}.$$



where R is the radius of the circle and r is the distance of the mass to the center of the force. The equation of motion is

$$F(r)\cos\theta = m\frac{v^2}{R}.$$

The angular momentum with respect to the center of the force is conserved:

$$L = m|\vec{r}\times\vec{v}| = mrv\cos\theta = \text{Const.}$$

Then $v = \dfrac{L}{mr\cos\theta}$. Thus

$$F(r) = m\frac{v^2}{R\cos\theta} = \frac{L^2}{mRr^2\cos^3\theta} = \frac{2L^2}{mr^3\cos^2\theta} = \frac{2L^2}{mr^3\left(\frac{r}{2R}\right)^2} = \frac{8R^2L^2}{mr^5}.$$

Solution3. The force law can also be determined by using the Binet equation

$$F(u) = -mh^2u^2\left(\frac{d^2u}{d\theta^2} + u\right)$$

where $u = \dfrac{1}{r}$ (the reciprocal of r) and $h = \dfrac{L}{m}$.

From $u = \dfrac{1}{2R\cos\theta}$, we have

$$\frac{du}{d\theta} = \frac{1}{2R}\frac{\sin\theta}{\cos^2\theta}, \quad \frac{d^2u}{d\theta^2} = \frac{1}{2R}\frac{\cos\theta\cdot\cos^2\theta - \sin\theta[2\cos\theta(-\sin\theta)]}{\cos^4\theta} = \frac{1}{2R}\frac{1+\sin^2\theta}{\cos^3\theta}$$

Thus

$$F=-mh^2u^2\left(\frac{1+\sin^2\theta}{2R\cos^3\theta}+\frac{1}{2R\cos\theta}\right)=-mh^2u^2\,\frac{1}{R\cos^3\theta}=-mh^2u^2\,\frac{8R^2}{(2R\cos\theta)^3}$$

$$=-8mh^2R^2u^5=-\frac{8mh^2R^2}{r^5}.$$

(The derivation of the Binet equation is as follows: The Newton's second law for the central force is

$$F(r)=m\left(\ddot{r}-r\dot{\theta}^2\right)$$

and the conservation of angular momentum can be written as

$$r^2\dot{\theta}=\frac{L}{m}=h=\text{constant}\,.$$

Then

$$\frac{du}{d\theta}=\frac{d}{dt}\left(\frac{1}{r}\right)\frac{dt}{d\theta}=-\frac{\dot{r}}{r^2\dot{\theta}}=-\frac{\dot{r}}{h},$$

$$\frac{d^2u}{d\theta^2}=-\frac{1}{h}\frac{d\dot{r}}{dt}\frac{dt}{d\theta}=-\frac{\ddot{r}}{h\dot{\theta}}=-\frac{\ddot{r}}{h^2u^2}.$$

Finally $F(r)=m\left(\ddot{r}-r\dot{\theta}^2\right)=-m\left(h^2u^2\frac{d^2u}{d\theta^2}+h^2u^3\right)=-mh^2u^2\left(\frac{d^2u}{d\theta^2}+u\right).$)))

4. A satellite is moving in a circular orbit of radius $2R_E$ about the Earth, where R_E is the radius of the Earth. The direction of motion of the satellite is suddenly changed through an angle α towards the Earth, without a change in speed. Find the angle α with which the trajectory of the satellite just touches the Earth.

Solution: When the A satellite is moving in a circular orbit of radius $2R_E$ about the Earth, its speed is determined by the equation

$$G\frac{M_E m}{(2R_E)^2}=m\frac{v^2}{2R_E}$$

where M_E is the mass of the Earth. Thus $v=\sqrt{\frac{GM_E}{2R_E}}$ and it energy is

$$E=\frac{1}{2}mv^2-G\frac{mM_E}{2R_E}=-G\frac{mM_E}{4R_E}.$$

After the direction is changed, the angular momentum of the satellite is

$$L=\left|m\boldsymbol{r}\times\dot{\boldsymbol{r}}\right|=m(2R_E)v\sin\left(\frac{\pi}{2}+\alpha\right)=m\sqrt{2GM_ER_E}\cos\alpha\,.$$

The condition for the satellite to just touch the Earth is that the perigee distance is R_E. Thus the conservation of energy gives

$$\frac{1}{2}mv_p^2-G\frac{mM_M}{R_E}=-G\frac{mM_E}{4R_E}\qquad\text{or}\qquad v_p=\sqrt{\frac{3}{2}\frac{GM_E}{R_E}}$$

where v_p is the speed of the satellite when it touches the Earth. The conservation of the angular momentum gives

$$mR_Ev_p=m\sqrt{\frac{3}{2}GM_ER_E}=m\sqrt{2GM_ER_E}\cos\alpha\,.\qquad\text{Therefore,}\quad\cos\alpha=\frac{\sqrt{3}}{2}\quad\text{and}\quad\alpha=\frac{\pi}{6}.$$